

MEMORANDUM

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2
Kopiereg voorbehou

Semester Test 2
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Module EBN111
Elektrisiteit en Elektronika
25 April 2012

Module EBN111
Electricity and Electronics
25 April 2012



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute 90 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	14		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 JD le Roux

Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.

Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	
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VRAAG/QUESTION 1 [24]

Vraag 1.1[4]

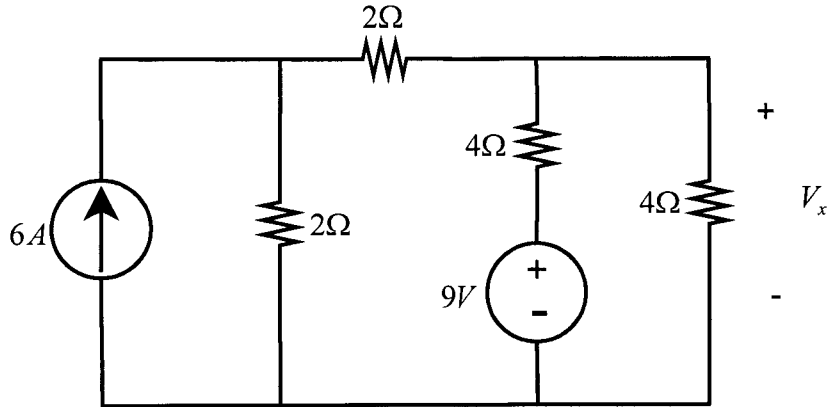
Gebruik superposisie:

- (a) Bereken V_x agv die 9V spanningsbron.
(b) Bereken V_x agv die 6A stroombron.

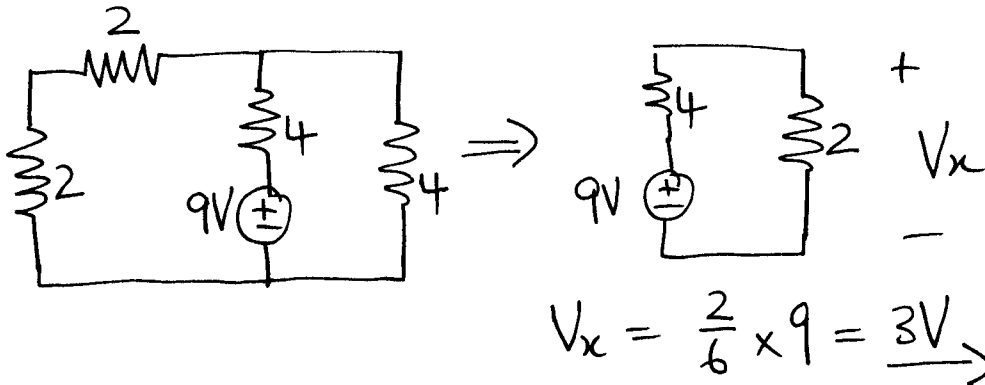
Question 1.1[4]

Use superposition:

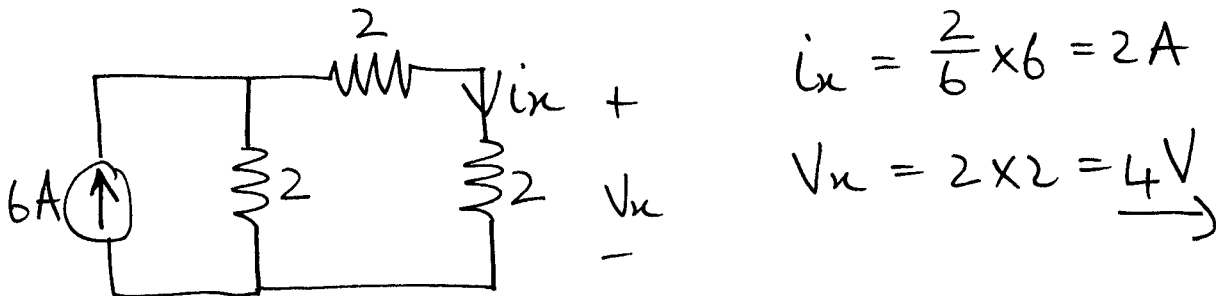
- (a) Determine V_x due to the 9V voltage source.
(b) Determine V_x due to the 6A current source.



(a)

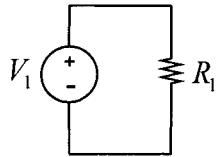


(b)

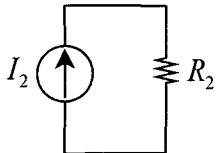


Vraag 1.2[4]

(a) Gebruik brontransformasies om die stroombaan in Fig. 2 te vereenvoudig na:

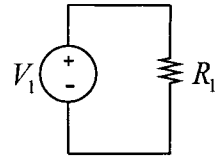


(b) Gebruik brontransformasies om die stroombaan in Fig. 2 te vereenvoudig na:



Question 1.2[4]

(a) Use source transformations to simplify the circuit in Fig. 2 to:



(b) Use source transformations to simplify the circuit in Fig. 2 to:

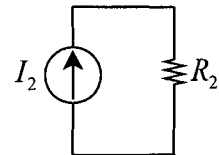
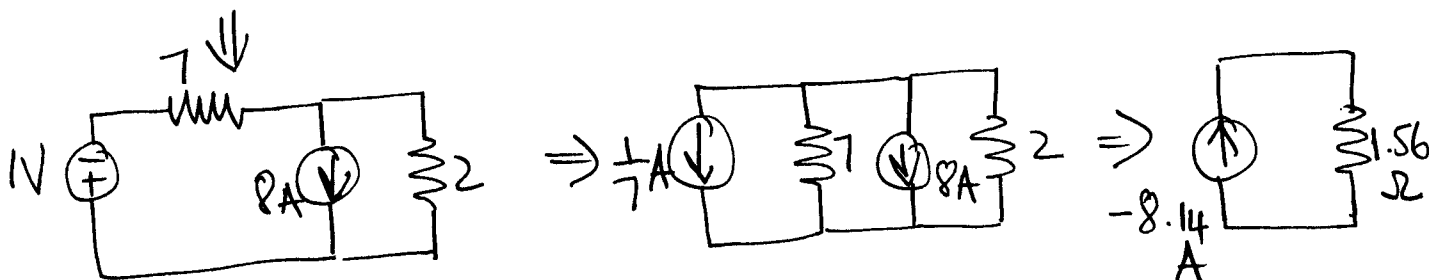
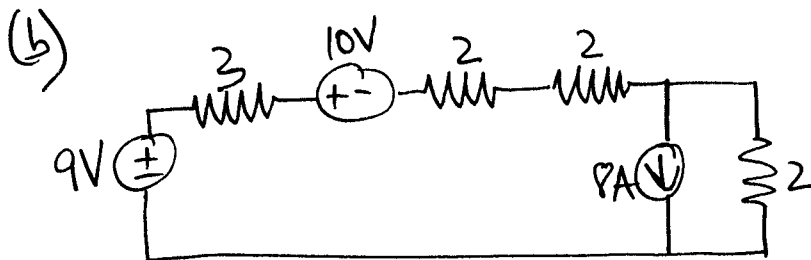
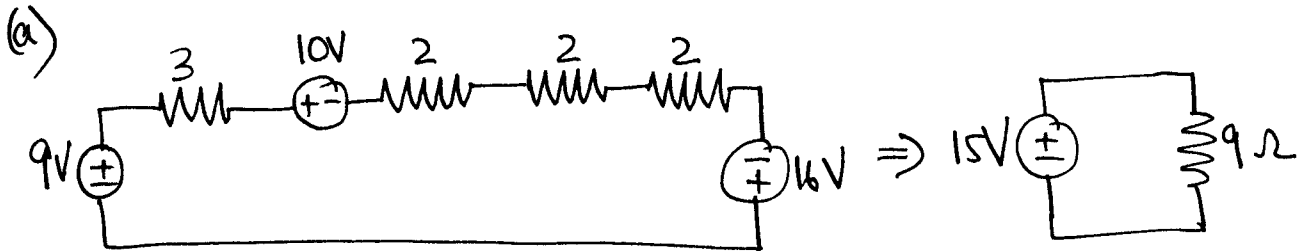
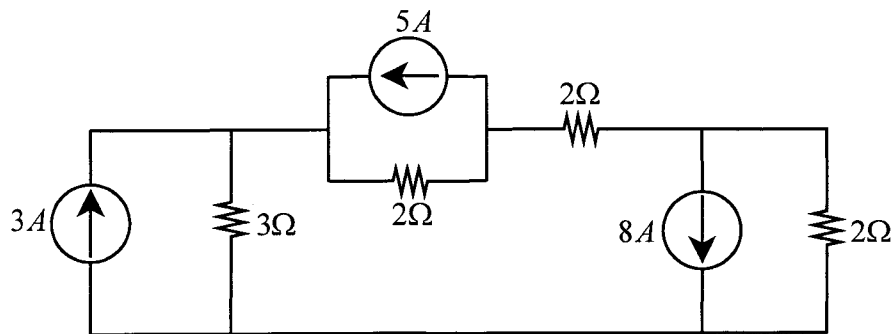


Fig. 2

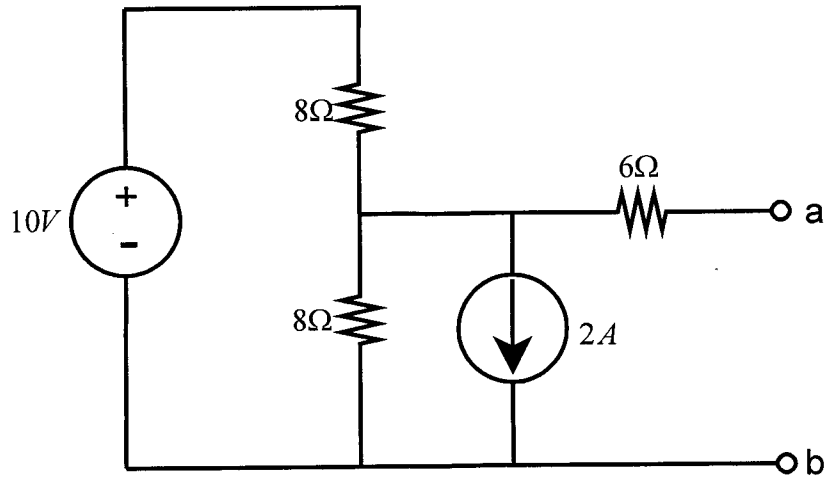
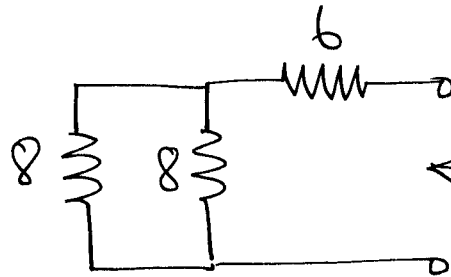


Vraag 1.3[4]

Bepaal die Thevenin ekwivalent tov terminale a-b.

Question 1.3[4]

Determine the Thevenin equivalent wrt terminals a-b.

 R_{th} 

$$R_{th} = 4 + 6 = 10\Omega$$

 V_{th}

Superposition:

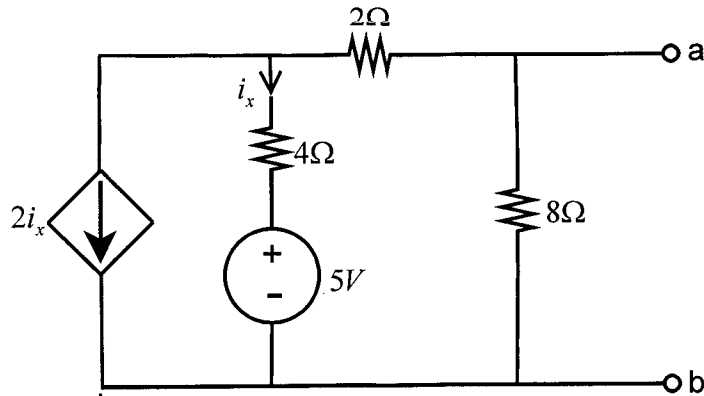
$$V_{th} = 10 \cdot \frac{8}{16} - 8 = -3V$$

Vraag 1.4 [4]

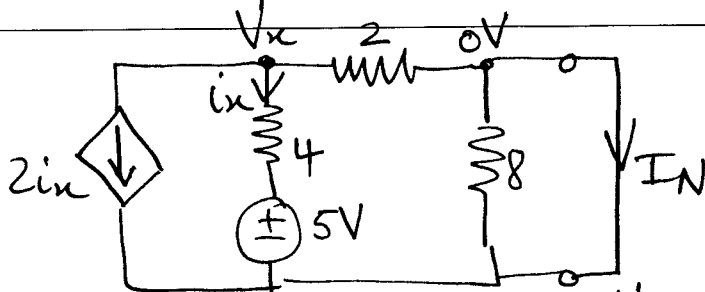
Bepaal die Norton ekwivalent tov terminale a-b.

Question 1.4 [4]

Determine the Norton equivalent wrt terminals a-b.



I_N :



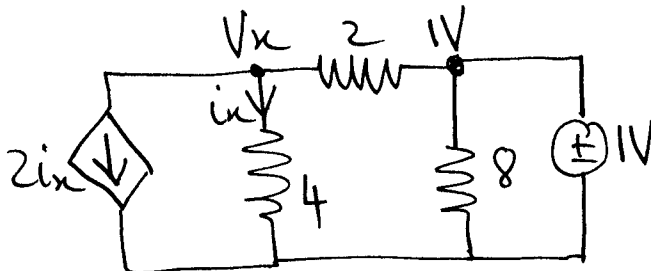
$$2i_x + i_x + \frac{V_N}{2} = 0 \quad i_x = \frac{V_N - 5}{4}$$

$$3\left(\frac{V_N - 5}{4}\right) + \frac{V_N}{2} = 0 \Rightarrow 3V_N - 15 + 2V_N = 0$$

$$V_N = 3V$$

$$I_N = \frac{V_N}{2} = 1.5A \rightarrow$$

R_N :



$$3i_x + \frac{V_N - 1}{2} = 0 \quad (V_N = i_x \cdot 4)$$

$$3\left(\frac{V_N}{4}\right) + \frac{V_N - 1}{2} = 0$$

$$3V_N + 2V_N - 2 = 0$$

$$V_N = 0.4V$$

$$I_0 = \frac{1}{8} + \frac{1 - 0.4}{2}$$

$$= 0.425A$$

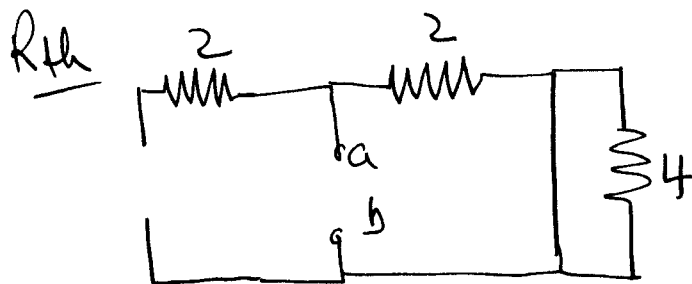
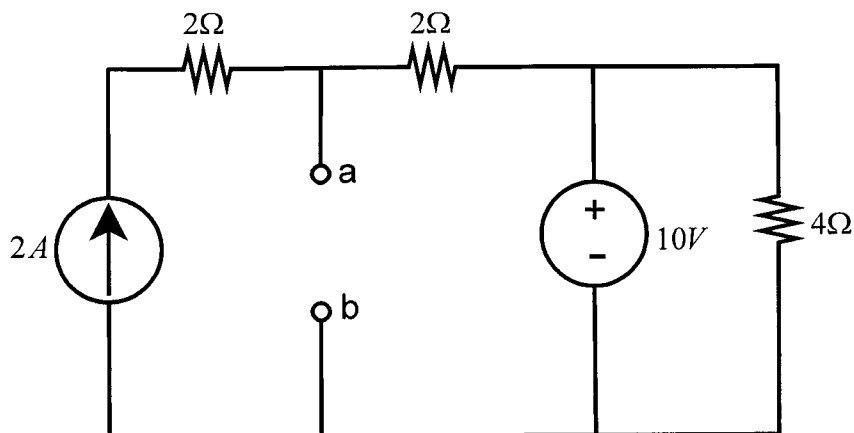
$$R_N = \frac{1}{0.425} = 2.35\Omega \rightarrow$$

Vraag 1.5[4]

Bepaal die Thevenin ekwivalent tov terminale a-b.

Question 1.5[4]

Determine the Thevenin equivalent wrt terminals a-b.



$$R_{th} = 2\Omega \rightarrow$$

V_{th} Superposition

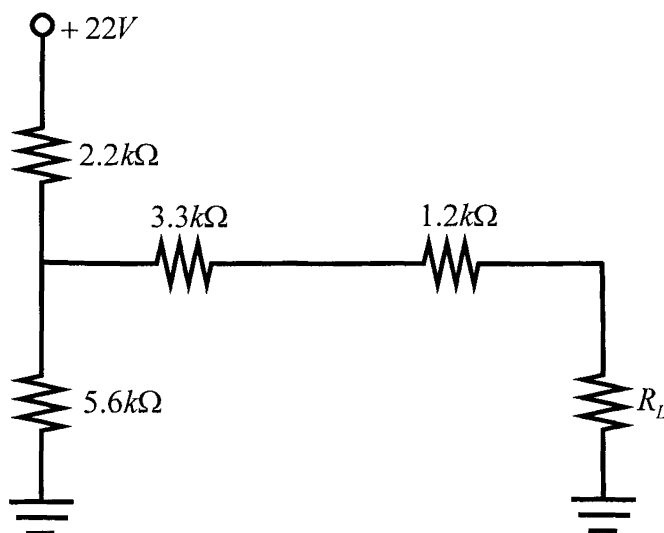
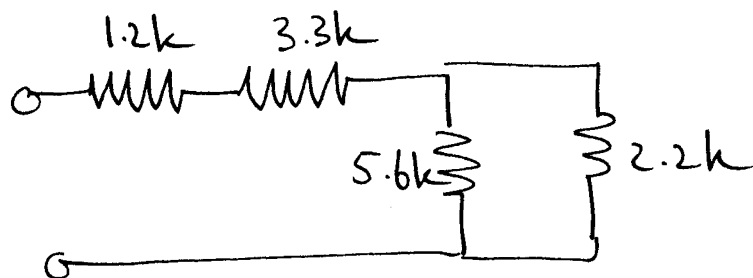
$$V_{th} = 2 \times 2 + 10 = 14V \rightarrow$$

Vraag 1.6[4]

- (a) Bepaal die waarde van R_L vir maksimum drywingsoordrag na R_L .
 (b) Bepaal die maksimum drywing na R_L .
 (Alle spannings is relatief tot die grond)

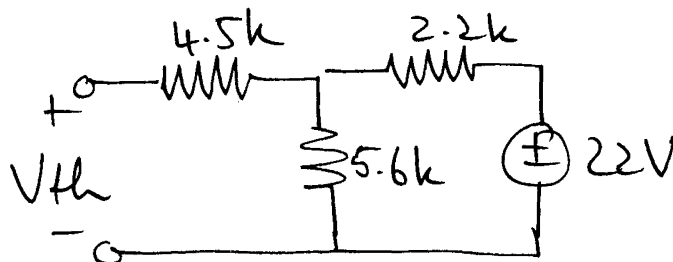
Question 1.6[4]

- (a) Find the value of R_L for maximum power transfer to R_L .
 (b) Find the maximum power to R_L .
 (All voltages are relative to the ground)

 R_{th} 

$$R_{th} = 1.2k + 3.3k + 5.6k // 2.2k$$

$$= 6.08k\Omega$$

 V_{th} 

$$V_{th} = \frac{5.6}{5.6 + 2.2} 22$$

$$= 15.8V$$

$$P_{max} = \frac{(15.8)^2}{4 \times 6.08k} = 10.26mW$$

EBN111 SEMESTER TOETS/TEST 2 (25 APRIL 2012)
ANTWOORDBLAD / ANSWER SHEET

Student se besonderhede/Student's details									
Van en voorletters: <i>Surname and initials:</i>					Studentenommer: <i>Student number:</i>				
Graadkursus: <i>Degree Course:</i>									
Taalgroep: <i>Language group:</i>	Afrikaans				English				
Tel. nr.: <i>Tel no.:</i>									

Q1		Q2		Q3		TOT:	/45
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VRAAG 1 – QUESTION 1 (24)		
1.1	$V_{x(9V)} = 3V$ ✓✓	$V_{x(6A)} = 4V$ ✓✓
1.2(a)	$V_I = 15V$ ✓✓	$R_I = 9\Omega$ ✓✓
1.2(b)	$I_2 = -8.14 A$ ✓	$R_2 = 1.56\Omega$ ✓
1.3	$V_{th} = -3V$ ✓✓	$R_{th} = 10\Omega$ ✓✓
1.4	$I_N = 1.5 A$ ✓✓	$R_N = 2.35\Omega$ ✓✓
1.5	$V_{th} = 14V$ ✓✓	$R_{th} = 2\Omega$ ✓✓
1.6	$R_L = 6.08 k\Omega$ ✓✓	$P_{max} = 10.26 mW$ ✓✓